

Recall from last class:

- A matrix is in **echelon form** if the following 3 properties hold:
 1. All non-zero rows are above zero rows.
 2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- A matrix in echelon form is in **reduced echelon form** if
 1. The leading entries are all 1.
 2. Each leading 1 is the only non-zero entry in its column.
- Each matrix is row equivalent to one and only one reduced echelon matrix.
- A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced row echelon form. A **pivot column** is a column that has a pivot position.
- **Row Reduction Algorithm (Gaussian Elimination).**
 1. The leftmost non-zero column is a pivot column.
 2. Select a non-zero entry in the pivot column as pivot. Interchange rows to move this entry to the top.
 3. Use elementary row operations to create zeros below the pivot.
 4. Apply 1-3 to the submatrix obtained by removing pivot row and column. Repeat until all non-zero rows are modified.
 5. Beginning with rightmost pivot and working upwards and to the left, create 0's above each pivot. Scale pivot positions entries to be 1.
- The variables corresponding to the pivot columns in an augmented matrix are called **basic** and the others are called **free**.
- A linear system is **consistent** if and only if the rightmost column in the augmented matrix is not a pivot column. If the system is consistent, then it has a unique solution when there are no free variables and infinite solutions when there is one or more free variables.
- A matrix with 1 column is called a **column vector**. A matrix with one row is called a **row vector**.
- We can add vectors of the same size by adding the corresponding components; namely, if
$$\vec{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} \text{ and } \vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \text{ then } \vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}.$$

Similarly, we can scale a vector by a real number c , by multiplying all the components by c ; namely, $c\vec{u} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}.$

- Given $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$ and scalars $c_1, \dots, c_p$, the vector

$$\vec{y} = c_1\vec{v}_1 + \dots c_p\vec{v}_p$$

is called the **linear combination** of $\vec{v}_1, \dots, \vec{v}_p$ with weights $c_1, \dots, c_p$.

- A vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \dots + x_n\vec{a}_n = \vec{b}$$

has the same solutions as the augment matrix

$$[\vec{a}_1 \ \vec{a}_2 \ \dots \ \vec{a}_n \left| \vec{b} \right].$$

Hence $\vec{b}$ is a linear combination of $\vec{a}_1, \dots, \vec{a}_n$ if and only if there is a solution to this matrix.

- If $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$, then the set of all linear combinations of $\vec{v}_1, \dots, \vec{v}_p$ is called the **span** of $\vec{v}_1, \dots, \vec{v}_p$. We write this set as $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$.

Let A be an $m \times n$ matrix, then

$$A = [\vec{a}_1 \ \dots \ \vec{a}_n], \quad \vec{a}_j \in \mathbb{R}^m, \ 1 \leq j \leq n.$$

If $x \in \mathbb{R}^n$, then we define

$$A\vec{x} = \vec{a}_1x_1 + \vec{a}_2x_2 + \dots + \vec{a}_nx_n.$$

Thus the matrix equation $A\vec{x} = \vec{b}$ is the same as the vector equation

$$\vec{a}_1x_1 + \vec{a}_2x_2 + \dots + \vec{a}_nx_n = \vec{b}.$$

1. Given

$$A = \begin{bmatrix} 1 & 3 & -4 \\ 1 & 5 & 2 \\ -3 & -7 & 6 \end{bmatrix} \quad \text{and} \quad \vec{b} = \begin{bmatrix} -2 \\ 4 \\ 12 \end{bmatrix}.$$

Is $A\vec{x} = \vec{b}$ consistent?

Fact: $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b} \in \text{Span}\{\vec{a}_1, \dots, \vec{a}_n\}$ where $\vec{a}_j$ is the j -th column of A .

2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Is $A\vec{x} = \vec{b}$ consistent for all $\vec{b} \in \mathbb{R}^3$.

Definition 1. A set of vectors $\vec{v}_1, \dots, \vec{v}_p$ in $\mathbb{R}^m$ **spans** $\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\vec{v}_1, \dots, \vec{v}_p$.

Theorem Let A be an $m \times n$ matrix, then the following are equivalent.

- (a) For each $\vec{b} \in \mathbb{R}^m$, the matrix equation $A\vec{x} = \vec{b}$ is consistent.
- (b) Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A .
- (c) The columns of A span $\mathbb{R}^m$.
- (d) A has a pivot position in every row.

3. Prove the theorem.

Fact (or Theorem 5 from the book): If A is a $m \times n$ matrix and $\vec{u}, \vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$, then

(a) $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$

(b) $A(c\vec{u}) = c(A\vec{u})$.

4. Prove the result.

5. **True or False.** If the augmented matrix has a pivot position in each row, then it is consistent.

6. Let

$$\vec{u} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} \text{ and } A = \begin{bmatrix} 2 & -5 \\ 2 & 6 \\ 1 & 1 \end{bmatrix}.$$

Is $\vec{u}$ in the span of the columns of A ?

Matrix Equations and Solutions – Answers / Solutions

1. An echelon form for $[A|b]$ is:

$$\left[\begin{array}{ccc|c} 1 & 3 & -4 & -2 \\ 0 & 2 & 6 & 6 \\ 0 & 0 & -12 & 0 \end{array} \right]$$

and the last column is not a pivot, so the system is consistent.

2. An echelon form for $[A|b]$ is:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & b_1 \\ 0 & -3 & -6 & b_2 - 4b_1 \\ 0 & 0 & 0 & b_3 - 2b_2 + b_1 \end{array} \right]$$

where $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in \mathbb{R}^3$ is an arbitrary vector. The last column is a pivot if $b_3 - 2b_2 + b_1 \neq 0$, so the system is consistent if $b_3 - 2b_2 + b_1 = 0$.

3. (a) $\implies$ (b): Suppose that (a) is true, and let $\vec{b} \in \mathbb{R}^m$. Then since $A\vec{x} = \vec{b}$ is consistent,

there is at least one solution $x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$. Let $\vec{a}_i$ be the i^{th} column of A . Then by definition

of matrix multiplication, $\vec{b} = \sum_{i=1}^n x_i \vec{a}_i$. Since $\vec{b}$ was arbitrary, any vector in $\mathbb{R}^m$ is a linear combination of the columns of A .

(b) $\implies$ (c): Suppose that (b) is true. In order for the columns of A to span $\mathbb{R}^m$, then any $\vec{b} \in \mathbb{R}^m$ must be a linear combination of the columns of A . This is exactly what (b) says.

(c) $\implies$ (d): Suppose that (d) is false, i.e. that A does not have a pivot position in every row, and let $\vec{b} \in \mathbb{R}^m$ be an arbitrary vector. Then by the row reduction algorithm, $[A|\vec{b}]$ is row equivalent to an augmented matrix $[B|\vec{c}]$ where B has at least one zero row at the bottom.

If the last entry of $\vec{c}$ is not zero, then the last column of $[B|\vec{c}]$ will be a pivot, so will not be consistent. Since row equivalent matrices have the same solution set, then $[A|\vec{b}]$ is also inconsistent, so there is no solution to $A\vec{x} = \vec{b}$, so $\vec{b}$ is not in the span of the columns of A .

Since row operations are reversible, just pick a $\vec{c}$ with the last entry nonzero, and then undo the row operations to get $\vec{b}$.

(d) $\implies$ (a): Suppose (d) is true, and let $\vec{b} \in \mathbb{R}^m$ be arbitrary. Form the augmented system $[A|\vec{b}]$ and perform row reduction. By the assumption that A has a pivot position in every row, then we will never encounter the situation where the last column is a pivot column. This is because to get the last column as a pivot column, we need to be able to perform row operations to get a zero row in A , which cannot happen by our assumptions. Thus the system $A\vec{x} = \vec{b}$ will always be consistent.

4. (a) Let $A, \vec{u}, \vec{v}$ be as follows:

$$A = [\vec{a}_1 \ \cdots \ \vec{a}_n] \quad \vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

Then we compute using the definitions:

$$\begin{aligned} A(\vec{u} + \vec{v}) &= A \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ \vdots \\ u_n + v_n \end{bmatrix} = (u_1 + v_1)\vec{a}_1 + \cdots + (u_n + v_n)\vec{a}_n \\ &= u_1\vec{a}_1 + v_1\vec{a}_1 + \cdots + u_n\vec{a}_n + v_n\vec{a}_n \\ &= (u_1\vec{a}_1 + \cdots + u_n\vec{a}_n) + (v_1\vec{a}_1 + \cdots + v_n\vec{a}_n) \\ &= A\vec{u} + A\vec{v}. \end{aligned}$$

(b) Exercise.

5. This is **false**. Consider the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 23 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

There is a pivot in each row, but the matrix is not consistent. A true statement here is that if augmented matrix does not have the last column as a pivot column, then it is consistent.

6. Remember that $\vec{u}$ being in the span of the columns of A is equivalent to the augmented matrix $[A|\vec{u}]$ being consistent. An echelon form for $[A|\vec{u}]$ is:

$$\left[\begin{array}{cc|c} 2 & -5 & 0 \\ 0 & 11 & 4 \\ 0 & 0 & \frac{30}{11} \end{array} \right]$$

so this system is not consistent, so $\vec{u}$ is not in the span of the columns of A .